Rotem Assouline

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RESEARCH INTERESTS

Interrelations of Riemannian geometry, convex geometry and optimal transport. Geometric and functional inequalities on weighted manifolds. Optimal transport with Lagrangian cost.

ACADEMIC POSITIONS

C.L.E Moore Instructor September 2026 – August 2029
Department of Mathematics, MIT
Mentor: Prof. Alexandr Logunov

FSMP Postdoctoral fellow October 2025 – September 2026
Institut de Mathématiques de Jussieu (IMJ-PRG), Sorbonne Université
Mentor: Prof. Dario Cordero-Erausquin

EDUCATION

Ph.D Mathematics 2025
Weizmann Institute of Science
Thesis: *Curvature-dimension for Autonomous Lagrangians.*
Advisor: Prof. Bo'az Klartag

M.Sc Mathematics 2020
Weizmann Institute of Science
Thesis: *Isometric Embeddings into Riemannian Surfaces with Hölder Continuous Curvature.*
Advisor: Prof. Bo'az Klartag

B.Sc Mathematics, magna cum laude, GPA 96 2018
Hebrew University

B.Mus Music, GPA 96 2017
Jerusalem Academy of Music and Dance

PUBLICATIONS

1. *Absence of free-time minimizers for N-body problems and a splitting theorem*, with Marco Mazzucchelli, preprint. arXiv:2610.00959
2. *Klingenberg's theorem on Anosov geodesic flows*, with Marco Mazzucchelli, preprint. arXiv:2608.09668
3. *Illumination bodies on Riemannian manifolds*, with Carsten Schütt and Elisabeth M. Werner, preprint. arXiv:2606.21112
4. *Magnetic Brunn-Minkowski inequalities*, preprint. arXiv:2606.08626
5. *Illumination Bodies in Projective Geometries*, with Florian Besau and Elisabeth Werner. arXiv:2605.25122
6. *Curvature-Dimension for Autonomous Lagrangians*, accepted to Geometric Aspects of Functional Analysis Israel Seminar (GAFA). arXiv:2409.08001
7. *Brunn–Minkowski Inequalities for Sprays on Surfaces.* Journal of Geometric Analysis 34, 339 (2024).

8. *A refinement of the Šidák-Khatri inequality and a strong Gaussian correlation conjecture*, with Shay Sadovsky and Arnon Chor, accepted to Geometric Aspects of Functional Analysis Israel Seminar (GAFA). arXiv:2407.15684.
9. *A finiteness principle for distance functions on Riemannian surfaces with Hölder continuous curvature*, Israel Journal of Mathematics (2026).
10. *Horocyclic Brunn-Minkowski inequality*, with Bo'az Klartag, Advances in Mathematics 436 (2024), Paper No. 109381, 39 pp.
11. *Growth Competitions on Spherically-Symmetric Riemannian Manifolds*, Analysis and Geometry in Metric Spaces 10 (2022), no.1, 146–154.

TEACHING

Teaching Assistant

- Harmonic Analysis, Tel-Aviv University, Spring 2025
- Functional Analysis, Tel-Aviv University, Spring 2022

Instructor

- De-Shalit summer camp, Weizmann Institute of Science, 2019
- Preparatory school, Azrieli College, Jerusalem, Summer 2018
- Preparatory school, Hebrew University, Jerusalem, Summer 2017

AWARDS

- Rothschild Fellowship, 2025
- Oberwolfach Leibniz Graduate Student grant (OWLG), 2025
- Lady Anne Chain Memorial Prize for outstanding PhD students, 2024
- Susan Sapir Memorial Prize for excellent master's students, 2021
- David Lopatie fellows program for outstanding master's students, 2018

COMMUNITY WORK

- Member of organizing committee, Workshop in High-dimensional Phenomena and Convexity, Kibbutz-resort Nahsholim (Technion), Israel, June 2025. Together with Galyna Livshyts, Dan Mikulincer, Liran Rotem and Shay Sadovsky.
- Referee and review work: Advances in Mathematics, Tohoku Mathematical Journal, Foundations of Computational Mathematics (FoCM), Journal of the London Mathematical Society, AMS Mathematical reviews (MathSciNet).

INVITED TALKS

- *Displacement convexity in electromagnetic optimal transport on Lorentzian manifolds.*
  - A Day of Convexity & High-Dimensional Geometry, Tel Aviv University, October 2026.
  - Convexity Seminar, Courant Institute, New York University, October 2026.
- *Magnetic Brunn-Minkowski inequalities.*
  - GdT Transport Optimal, EDP et Machine Learning, Institut de mathématique d'Orsay, June 2026.
  - Séminaire d'Analyse, Institut de Mathématiques de Toulouse, May 2026. Slides
- *Curvature-Dimension for Autonomous Lagrangians.*

  - Geometric Structures Research Seminar, SISSA, April 2026. Slides
  - Functional Inequalities: Geometric Calculus meets Stochastic Analysis, Oberwolfach, December 2025.
  - Seminario di Analisi Matematica, Dipartimento di Matematica 'Federigo Enriques', Università degli Studi di Milano, October 2025. Slides
  - Dipartimento di Matematica e Informatica 'Ulisse Dini', Università degli Studi di Firenze. October 2025.
  - Seminario di Equazioni Differenziali, Dipartimento di Matematica, Università degli Studi di Roma 'Tor Vergata', September 2025.
  - Geometric Functional Analysis & Probability Seminar, Department of Mathematics, Weizmann Institute of Science, September 2025.
  - Workshop in High dimensional Phenomena and Convexity, Kibbutz-resort Nahsholim, June 2025.

- *Brunn-Minkowski inequalities for autonomous Hamiltonians on weighted manifolds.* Convex Geometry and its Applications, Oberwolfach, December 2024.
- *Magnetic Brunn-Minkowski and Borell-Brascamp-Lieb inequalities on Riemannian manifolds.*
  - OxPDE Lunchtime Seminar, Oxford, November 2024.
  - Analysis seminar, Georgia Tech, October 2024.
  - Online AGA Seminar (Zoom), September 2024. Video
- *Brunn-Minkowski inequalities for sprays on surfaces.*
  - High dimensional phenomena: geometric and probabilistic aspects, HIM, Bonn, March 2024. Video
  - Geometric Functional Analysis & Probability Seminar, Weizmann Institute, Rehovot, January 2024.
  - Geometry and Probability, Saitama, August 2023.
  - Geometric Inequalities, Convexity and Probability, BIRS-IMAG, Granada, June 2023.
  - Informal Geometric Analysis Seminar (Zoom), University of Maryland, February 2023.
- *Horocyclic Brunn-Minkowski inequality.*
  - Workshop on geometric analysis with focus on geometric flows, Hebrew University, Jerusalem, March 2023.
  - Probabilistic Methods in Geometry and Analysis, ICERM, Brown University, Providence, RI, October 2022. Video
- *Are there finiteness principles for Riemannian embeddings?* 14th Whitney Problems Workshop, 2021 (Zoom).